

Domain / Theme	Architecture	Population Reach	Scheme Catalog
E-Governance & Welfare	Decoupled Monorepo	1.4B Indian Citizens	4,770+ myScheme Portals

Zero PII

MASKED IDENTIFIERS

Page 1 of 4

3. Custom DigiLocker Vault & Demographic Personas

To maximize responsiveness on resource-constrained mobile devices and rural networks, Sahayak replaces heavy 3D WebGL canvases with a high-performance **2D React/TailwindCSS Custom DigiLocker Vault** dashboard.

Demographic Persona	Target Demographic Profile	Socioeconomic Factors	Matched Official Schemes
Small Farmer (Rameshwar Sharma)	Age 38, Male, Married, OBC, Uttar Pradesh	Income Rs. 1.6L/yr, 2.5 Acres Land, Crop Insured	PM-KISAN (Rs. 6,000/yr), PM Fasal Bima Yojana
College Student (Anjali Gupta)	Age 19, Female, Single, OBC, Bihar	Family Income Rs. 1.2L/yr, Undergrad (B.Sc 2nd Yr)	National Scholarship Portal (NSP Post-Matric Grant)
Girl Child / Guardian (Sunita & Priya)	Age 34 (Mother), General, Rajasthan. Child Age 8	Income Rs. 1.4L/yr, Natural Guardian, Small Savings	Sukanya Samriddhi Yojana (8.2% Sovereign Interest)
Widow Destitute (Kamla Devi)	Age 52, Female, Widow, SC, Madhya Pradesh	Income Rs. 72k/yr, BPL Card holder, Unemployed	Indira Gandhi Widow Pension (IGNWPS / NSAP)
Senior Citizen (Ramprasad Verma)	Age 67, Male, Married, OBC, Uttar Pradesh	Income Rs. 84k/yr, BPL Card holder, Retired	Indira Gandhi Old Age Pension (IGNOAPS / NSAP)
Woman Entrepreneur (Pooja Meena)	Age 29, Female, Single, ST, Rajasthan	Income Rs. 2.1L/yr, Greenfield Micro-Enterprise	Stand-Up India (Rs. 10L-1Cr), PM Mudra Yojana

4. Comprehensive Official Scheme Catalog (myScheme.gov.in Mapped)

All schemes in the platform are classified into official **myScheme.gov.in categories** and link directly to verified sovereign .gov.in and .nic.in portals:

Official myScheme Category	Scheme Title	Verified Portal URL	Key Evaluated Criteria
Agriculture, Rural & Environment	PM-Kisan Samman Nidhi	https://pmkisan.gov.in/	Farmer, landholder, income < Rs. 4 Lakhs
Agriculture, Rural & Environment	PM Fasal Bima Yojana (PMFBY)	https://pmfby.gov.in/	Landholding farmer, crop loss insurance
Education & Learning	National Scholarship Portal	https://scholarships.gov.in/	Enrolled student, family income < Rs. 2.5L
Women and Child	Sukanya Samriddhi Yojana	https://www.indiapost.gov.in/	Girl child age ≤ 10 yrs, natural guardian
Women and Child	PM Matru Vandana Yojana	https://pmmvy.wcd.gov.in/	Pregnant or lactating mother, age ≥ 19 yrs
Social Welfare & Empowerment	Indira Gandhi Widow Pension	https://nsap.nic.in/	Female, widow status, age ≥ 40 yrs, BPL
Social Welfare & Empowerment	Indira Gandhi Old Age Pension	https://nsap.nic.in/	Senior citizen age ≥ 60 yrs, BPL cardholder
Health & Wellness	Ayushman Bharat (PM-JAY)	https://nha.gov.in/	Cashless Rs. 5L cover, family income ≤ Rs. 5L / BPL
Business & Entrepreneurship	Stand-Up India Scheme	https://www.standupmitra.in/	SC/ST or Women Entrepreneurs (Rs. 10L–1Cr)
Business & Entrepreneurship	PM Mudra Yojana (Shishu)	https://www.udyamimitra.in/	Self-employed, micro-artisan credit ≤ Rs. 50k
Housing & Shelter	PMAY - Gramin	https://pmayg.nic.in/	Rural household, no pucca house, income ≤ Rs. 3L
Universal Discovery	myScheme Universal Platform	https://www.myscheme.gov.in/	Discovery index for 4,770+ Central/State schemes

5. High-Performance Rust Rule Engine & trait SchemeMatcher

The core matching engine is written in **Rust 2021** using the asynchronous **Axum** framework. It processes citizen profiles concurrently against thousands of rules with zero garbage collection pauses. In backend/src/models/scheme.rs, the trait SchemeMatcher evaluates 14 criteria dimensions:

- **14 Evaluated Dimensions:** Minimum Age, Maximum Age, Senior Citizen (≥60y), Gender, Marital Status (Widow), Stand-Up India (SC/ST/Women), Girl Child Eligibility (≤10y), Maternal Health/Pregnancy, BPL/Antyodaya Status, Income Ceilings, Social Category (General/OBC/SC/ST/EWS), Landholding Acreage, Active Student Enrollment, and Differently Abled (PwD).

```
pub trait SchemeMatcher {
    fn evaluate(&self; citizen: &CitizenProfile; lang: &str;) -> SchemeMatchResult;
}

impl SchemeMatcher for Scheme {
    fn evaluate(&self; citizen: &CitizenProfile; lang: &str;) -> SchemeMatchResult {
        let mut satisfied_rules = Vec::new();
        let mut unmet_rules = Vec::new();
        let mut score: u32 = 100;
        // High-concurrency evaluation across 14 demographic dimensions...
        SchemeMatchResult { scheme: self.clone(), is_eligible, match_score: score, ... }
    }
}
```

6. Multilingual Bhashini Pipeline & Chrome MV3 Auto-Fill

A. End-to-End Language Mirroring (Bhashini Architecture)

Sahayak implements dynamic language mirroring: when a user inputs audio or text in Hindi (or Tamil, Telugu, Bengali, Marathi), the platform detects the script, matches eligibility, and generates both text responses and synthesized TTS audio in the **exact matching Indian language**. Custom typography loaded from Google Fonts (Noto Sans Devanagari, Rozha One) ensures crisp ligature rendering.

B. Chrome Manifest V3 Auto-Fill Engine

- **Fuzzy DOM Matching:** Employs normalized Levenshtein string distance and semantic synonym dictionaries to discover fields across disparate state portals (matching labels, IDs, names, and ARIA attributes for Aadhaar, income, land, and bank details).
- **Synthetic Event Dispatch:** Dispatches bubbling input, change, and blur events to guarantee that modern reactive frameworks (React, Angular, Vue, and legacy ASP.NET) acknowledge injected values.
- **Cross-Context Bridge:** Synchronizes profile data across the Web App and Extension via extensionBridge.js using window.postMessage and CustomEvents.

7. Strict Data Privacy & Zero Realistic PII Framework

To adhere strictly to Indian sovereign data governance policies and prevent accidental generation of valid personal credentials:

- **Aadhaar Numbers:** Strictly output as placeholder strings XXXX-XXXX-XXXX or [Aadhaar Redacted]. Zero realistic 12-digit numeric sequences are generated.
- **Mobile Numbers:** Formatted strictly as [Phone Redacted].
- **Certificate IDs:** Redacted via standardized placeholder masks (e.g. UP/REV/INC/XXXX/XXXXXX, DL-IND-XXXX-XXXXXX).

8. Technical Specifications & Dependencies

Subsystem	Core Technology & Version	Role & Key Packages
Frontend (PWA)	React 18.3, Vite 5.3, TailwindCSS 3.4	Custom DigiLocker 2D Vault, Zustand 4.5 store, Lucide React icons, Responsive CSS.
Backend (REST)	Rust 2021, Axum 0.7, Tokio 1.38	Concurrent trait SchemeMatcher, Serde 1.0, Tower-HTTP 0.5 CORS, Tracing logging.
Auto-Fill Engine	Chrome Manifest V3, Service Worker	Levenshtein DOM heuristics, Content Scripts, Event bubbling, storage permissions.
Voice Engine	Web Speech API, Bhashini AI Architecture	Multilingual ASR, SpeechSynthesis TTS, Noto Sans Devanagari typography.

9. REST API Endpoint Reference (Port 8080)

Method	Endpoint	Payload & Description
GET	/api/health	Health check returning service uptime and operational status.
GET	/api/schemes	Returns complete master list of schemes across official myScheme categories.
GET	/api/digilocker/documents	Returns simulated verified DigiLocker certificates with strict placeholder IDs.
POST	/api/match	Evaluates CitizenProfile against criteria rules; returns scored matches.
POST	/api/chat	Multilingual conversational assistant returning localized advice and suggestions.

10. Quick Start & Local Demonstration Guide

Follow these steps to run Sahayak in full offline/local demo mode:

```
# 1. Launch Rust Backend (Port 8080)
cd backend
cargo run

# 2. Launch React Frontend PWA (Port 5173)
cd frontend
npm run dev

# 3. Load Chrome MV3 Extension
- Open Chrome -> navigate to chrome://extensions/ -> Toggle 'Developer mode'
- Click 'Load unpacked' -> Select the 'extension' directory
- Open http://localhost:5173 and test 1-click auto-fill on http://localhost:5173/demo_portal.html
```